No. of Functional Features Included in the Solution

Date	14 July 2026
Team ID	-
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Components	Status (Done / In Progress / Pending)	Marks Contribution
1	Audio Upload	Upload voice/audio explanation for analysis	Upload Page, File Uploader	Done	High
2	Speech-to-Text	Converts speech into text using Faster Whisper	Whisper Model	Done	High
3	Audio Analysis	Calculates duration, speech rate, loudness and silence	Audio Analysis Module	Done	Medium
4	Semantic Similarity	Compares student explanation with reference concept	Sentence-BERT Model	Done	High
5	Keyword Matching	Detects matched and missing keywords	Keyword Matching Module	Done	Medium
6	AI Score Generation	Calculates concept understanding score	Scoring Module	Done	High
7	AI Feedback	Provides personalized feedback based on score	Feedback Module	Done	Medium
8	PDF Report Generation	Generates downloadable analysis report	PDF Generator	Done	High
9	Analysis History	Saves results in SQLite database	Database Module	Done	Medium

10	Dashboard	Displays reports and analysis history	Dashboard Page	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	10
Core / Must-Have Features	7
Additional / Nice-to-Have Features	3
Features Tested & Verified	10

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Home Page, Upload Page, Dashboard, Reports
2	Backend / Logic	4	Speech-to-Text, Semantic Similarity, Scoring, Feedback
3	Database / Storage	1	SQLite Analysis History
4	API / Integration	2	Faster Whisper, Sentence-BERT
5	Security / Authentication	0	Planned for Future Enhancement